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EDGE-AI BASED PREDICTIVE CLOSED CONTROL SYSTEM FOR SMART POULTRY FARMING

Minor Project Proposal Presentation

Project Team

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Motivation

- Poultry farms require continuous environmental monitoring.
- Manual monitoring is labor-intensive and reactive.
- Uncontrolled environmental levels affect birds' health and growth.
- Existing systems lack predictive intelligence.
- Need for autonomous and cost-effective solutions.

Objectives of Project

- 1 To design and develop an IoT-based Edge-AI system for real-time monitoring, prediction, and control of environmental conditions in poultry houses, with a focus on the brooding stage of broiler production.

Scope of Project

- Targets commercial broiler poultry farms.
- Real-time NH_3 and temperature data acquisition.
- Edge AI prediction using MLP on ESP32.
- Automated ventilation and alarm actuation on prediction.
- Web-based dashboard for remote farm monitoring.

Literature Review

Paper	Author	Focus	Dataset	Compute	Method	Results	Limitations (we address)
Surateno (2024) [4]	Surateno S. et al.	IoT closed house control	Real-time sensors	ESP32	Threshold-based	Automated control	Reactive only; no prediction
Cassuce (2013) [6]	Cassuce D. et al.	Broiler temperature needs	Controlled chambers	Statistical	Thermal comfort index	Week1:32-35°C	No adaptive real-time control
Miles (2004) [5]	Miles D. et al.	NH3 impact on growth	Controlled chambers	Statistical	Dose-response	12% loss at 50ppm	No prevention; retrospective only
Balthazar (2025) [7]	da Rocha Balthazar G. et al.	MQ-137 validation	Poultry house + lab	ESP8266	Sensor calibration	R=0.82, R ² =0.64	Monitoring-only; no ML integration
Barbosa (2024) [3]	Barbosa L. et al.	NH3 risk prediction	Commercial farm, 12 weeks	Desktop CPU	MLP vs DT vs NB vs KNN	MLP >98% accuracy	No edge-deployment; no real-time control

Gap Analysis

Limitations of Existing Solutions:

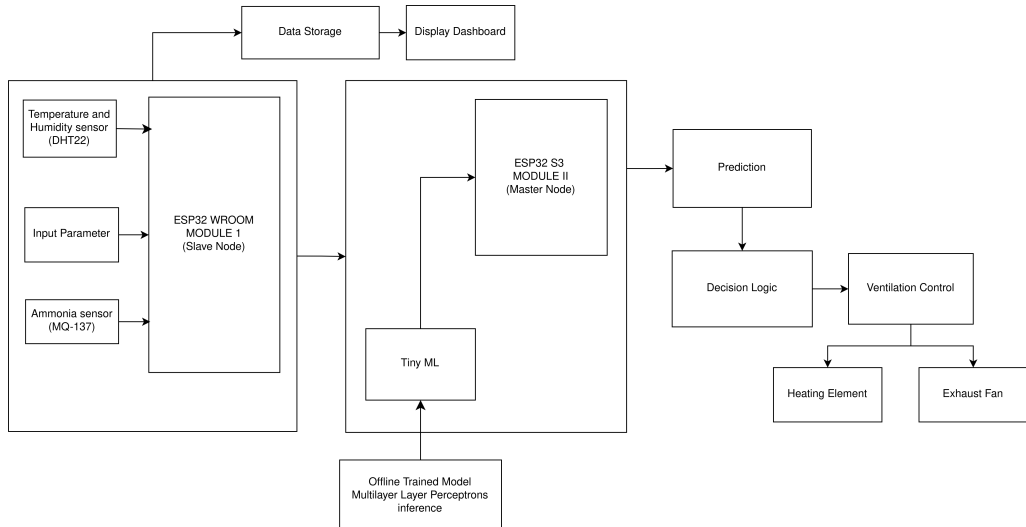
- **Reactive Control:** Surateno (2024) and similar IoT systems only respond *after* thresholds are breached, exposing chicks to harmful conditions.
- **No Edge-AI:** Barbosa (2024) achieved accurate MLP predictions but on desktop CPUs — no real-time, on-device inference.
- **No Closed-Loop Integration:** Balthazar (2025) validated MQ-137 but monitoring-only; no automatic actuation.
- **Age-Blind Control:** Most systems ignore broiler age, despite Cassuce (2013) proving temperature needs change weekly.
- **No Predictive Actuation:** No existing low-cost system combines **TinyML prediction + ESP-NOW communication + automatic ventilation/heating control**

Research Gap Addressed by Our Project

Our system integrates:

- **TinyML (MLP)** for *predictive* ammonia and temperature spike detection
- **ESP32-S3 edge deployment** for real-time inference (no cloud dependency)
- **Age-adaptive week-based thresholds** (Weeks 1-4+ per Cassuce 2013)
- **Closed-loop control** activating ventilation/heating *before* critical limits
- **ESP-NOW communication** for reliable farm-area networking

System Architecture



Input Layer — Data Acquisition

- MQ-137 sensor measures ammonia concentration in ppm.
- DHT22 sensor captures ambient temperature values.
- Data collected at fixed time intervals continuously.
- Normalized data passed to the processing layer.

Processing Layer — Prediction & Decision

- Sliding window input fed into MLP.
- MLP forecasts future NH_3 and temperature values.
- The model runs on ESP32 via TFLite Micro inference engine.
- All processing happens on-device.

Output Layer — Actuation & Monitoring

- Ventilation fan activated on Hazard detection.
- Heating element triggered on low temperature prediction.
- Web dashboard displays live and predicted sensor values.
- The operator receives real-time alert messages on hazards.
- Events logged for future farm data analysis.

Tools and Technologies

Programming Languages:

- Python
- C++
- JavaScript

Database:

- MySQL

Development Tools:

- VsCode
- git
- jupyter notebook

Hardware Implementation

- ESP32-WROOM & ESP32-S3
- DHT22
- MQ-137
- 2-Channel Relay
- DC Fan
- PTC Heating Element

Dataset

1	Sr No	Temperature	Humidity	CO2	NH3	Light Intensity
2	169	31	51	2655	27	19
3	170	27	68	2855	22	10
4	171	28	63	3025	24	13
5	172	29	57	3124	25	10
6	173	27	70	2926	29	12
7	174	30	54	3256	22	17
8	175	29	63	3175	22	17
9	176	28	62	3375	23	19
10	177	28	58	3025	27	16
11	178	32	52	2860	28	10
12	179	31	50	2684	27	20
13	180	29	58	3089	26	18
14	181	32	53	3223	26	13
15	182	27	65	2939	24	15
16	183	28	57	2672	28	20

Figure: Original dataset for MLP training

[1]

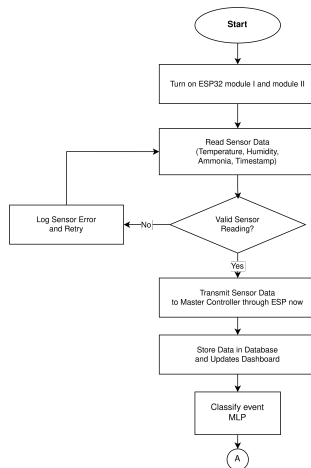
Dataset

	A	B	C	D	
1		humidity	temperature	ammonia	
2	0	63,72	21,3	10,9	
3	1	58,54	22,6	12,2	
4	2	62,3	23,2	11,8	
5	3	62,85	22,3	5,9	
6	4	63,35	21,8	5,6	
7	5	67,94	20,7	4,7	
8	6	62,1	22,7	7,3	
9	7	66,2	21,7	3,7	
10	8	65,86	22,2		9
11	9	63,41	22,7	6,7	
12	10	58,72	23,4	12,3	

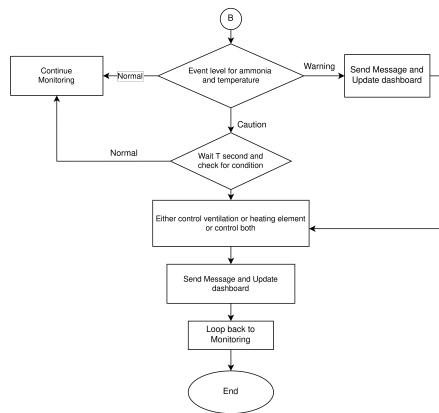
Figure: Additional dataset for MLP training

[2]

Flowchart



Flowchart



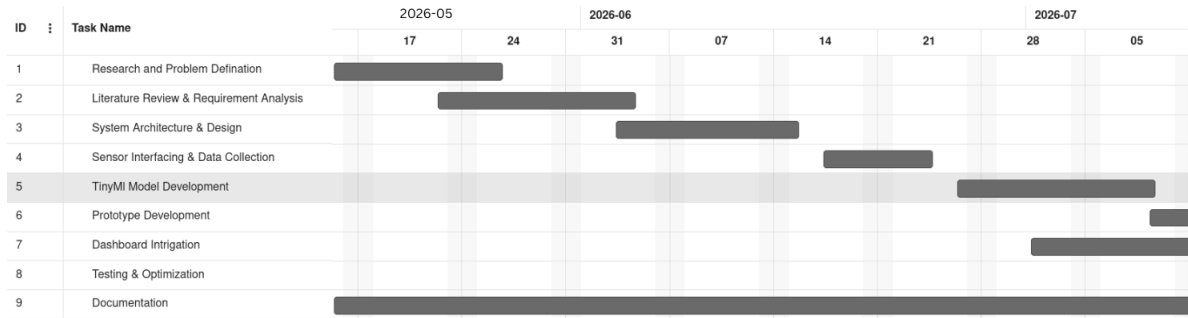
Expected Results

- MLP predicts ammonia and temperature.
- Hazard class identified before critical threshold is reached.
- Ventilation and alarm triggered automatically on hazard detection.
- Dashboard displays live sensor and prediction data accurately.
- Reduced manual monitoring effort for poultry farm operators.

Project Applications

- Continuous air quality monitoring in poultry farm houses.
- Early hazard detection to prevent bird mortality losses.
- Automated ventilation control for optimal farm environment.
- Scalable deployment across multiple poultry farm units.
- Data logging for farm productivity and environmental factors.
- Applicable to livestock and greenhouse monitoring systems.

Gantt Chart



Budget

Component Cost Breakdown

SN	Components	Qty	Unit Cost (NPR)	Total (NPR)
1	ESP32 S3	1	1500	1500
2	ESP32 Dev Board	1	750	750
3	DHT22 Sensor	1	350	350
4	MQ-137 Ammonia Sensor	1	2000	2000
5	2-Channel Relay Module	1	250	250
6	DC Cooling Fan	1	250	250
7	PTC Heating Element	1	350	350
8	Heat Sink	1	250	250
9	Power Supply Adapter	1	500	500
10	Voltage Regulator	1	150	150
11	Jumper Wires + Breadboard	1	300	300
12	Veroboard	1	60	60
13	Enclosure Box	1	300	300
			Total	7010 NPR

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-  A. Barsagade. (2023). *Poultry Farm Environmental Sensor Dataset*. IEEE Dataport. DOI: <https://doi.org/10.21227/dpkp-kx84>
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